X-Band Power Amplifier for Next Generation Networks Based on MESFET

Muhammad Saad Khan[1,4], Hongxin Zhang *[1,2], Fan Zhang [3], Sulman Shahzad[4], Rahat Ullah[5], Sajid Ali[6], Qasim Ali Arain[1], Manzoor Ahmed[7]

[1] School of Electronic Engineering, Beijing University of Posts and Telecommunications (BUPT), Beijing 100876, China
[2] Beijing Key Laboratory of Work Safety Intelligent Monitoring, Beijing University of Posts and Telecommunications, Beijing 100876, china
[3] College of Information Science and Electrical Engineering, Zhejiang University, Hangzhou, Zhejiang, 310027,china
[4] Electrical Engineering Department, University College of Engineering & Technology, Bahauddin Zakariya University, Multan, Pakistan
[5] State Key Laboratory of Information Photonics and Optical Communications, Beijing University of Posts and Telecommunications, Beijing 100876, China
[6] Department of Computer Science, D.G. Khan Campus. University of Education, Lahore, Pakistan
[7] Future Internet & Communication Lab (FIB), Department of Electronic Engineering, Tsinghua University, Beijing 100084, China
[*] The corresponding author, Email: hongxinzhang@bupt.edu.cn

Abstract: Advanced wireless standards of communication like 3GPP and LTE are becoming more and more efficient and with this evolution of communication systems mobile equipment is also become smaller and smaller. Power amplifier designing has become a very crucial task in this era where efficiency and size are the main concern of any designer. In this paper we have design and analyzed X-band Class E Metal-semiconductor field effect transistor (MESFET) based Power Amplifier. This device targets the devices which use OFDM technique to improve their spectral efficiency for the next generation communication systems. Microstrip lines are used to achieve small size for our design instead of lumped components. Load Pull measurements are used to get MESFET input and output impedances optimum values. For linear and non linear operation small signal mathematical model of the design is used. To reduce thermal losses FR4 substrate is used to increase PA efficiency. Our designs shows small values of input and output return loss of about -22.3dB and -23.716 dB achieving a high gain of about 25.6 dB respectively, with PAE of about 30 % having stability factor greater than 1 and 21.894dBm of output power.

Keywords: MESFET; X-band; Power Amplifier; Advance Design System; PAE; LTE

I. INTRODUCTION

Power amplifiers are basically designed by balancing the basic three requirements i.e. efficiency, linearity and bandwidth. But when PA is used in high power applications efficiency is the main concern for a designer. Class E,F are the most commonly used high efficiency PA for which 100 percent theoretical efficiencies may be achieved but at the cost of most common three basic factors i.e. linearity, gain and bandwidth [1][2]. Despite of their advantages in high efficiency category they have limited advantages when it comes to the high radio frequency applications such as in microwave communications where this system makes use of the large bandwidth sacrificing its signal strength [3]. The improvement in efficiency of power amplifiers reduces power consumption

Received: May 23, 2016
Revised: Aug. 09, 2016
Editor: Xinbing Wang

In order to improve their spectral efficiency for the next generation communication systems, the authors designed a Class E Power amplifier for X Band operation.

and thus diminishes the requirements for heat sink and in turns increases efficiency/performance. A power amplifier consumes major part of power in wireless communication. In order to improve efficiency class-E amplifier is used because of its switching operation, high efficiency at high frequency above 2GHz [4][5].Gallium Arsenide Field Effect transistor [GaAs] is very important device for X-band frequencies because of its stability and power efficiency for low power requirements. For power amplifier and oscillators design GaAs MESFET is an ideal device. In order to design a power amplifier in ADS its linear behavior should be checked first. Therefore various forms of analytical models have been developed in order to understand the behavior of MESFETs [6]. The most important things for FETs linear operation are its drain-source and gate-source voltage characteristics. The simulation results for S parameter which are frequency dependent and linear properties such as maximum gain, stability are affected by the charge model accuracy [7].

The main aim of this paper is to design and analyze PA which will operate in the portable next generation communication devices and will be used as solid-state transmitter in the terminal stage. The main approach and design is entirely dependent on S-parameters [8]. As we know that load and source terminations are complex impedances so stability is achieved by the conductance which results by the transformation of load and source terminations of the matching circuits [9]. Today the industry is focusing to use 5g communication systems, in future the X-band will be a valuable region of frequency interms for high speed wireless communications. In the era of high speed communication system, the design of a power amplifier is the most crucial thing. Although modern transistors like HEMT and PHEMT can provide far better results than MESFET but they are quite expensive [10]. In MESFETS devices the peak drain to source will occur when the gate voltage reaches about zero which means that these devices work in the depletion mode. The device current reaches to zero as the gate voltage becomes more and more negative. There are two basic parameters which should be kept in mind while designing a MESFET PA, these are the pinch-off voltage and the maximum current. For a system to provide steady performance the drain current of the system must be controlled to adjust the gain and PAE [11].

The next generation communication devices use polar modulators in which the bias tracking of power amplifier is instantaneous and controlled by modulating the supply voltage. The input signal is converted into envelope and phase signals. Further the phase signal goes to the input of power amplifier while the envelop signal moves to the supply of the power amplifier. Eventually, both phase and envelop information are combined at the power amplifier stage.

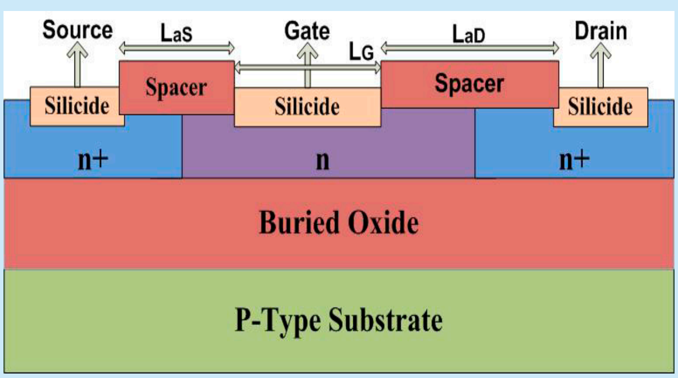

Fig. 1 *MESFET cross-section [12]*

II. MESFET ARCHITECTURE DESCRIPTION

As shown in fig.1 a partially depleted MESFET cross section is shown. An important step while designing a MESFET is incorporation of the silicide layer for making spacers between the source-gate and between gate-drain which also helps in preventing shorting, which may occur between its three terminals and thus giving high breakdown advantage to the MESFET [12][13].

In this process a Schottky gate was created by adding metal silicide step over n-well

Fig. 1 need vector diagrams, please redo it with font times new roman 6pt which is lightly doped ss shown in fig.1. The Gate length is *LG* which is the distance between the two spacers and *LaD* is spacers length to the drain and the access length is defined by *LaS*, the spacing and sizing of *LaD* and *LaS* are the most important parameters that defined the MESFETS performance [14].

High voltage characteristics of the MESFETs are due to its Schottky gate. Schottky gate can handles very high amount of current flow. MESFETs devices avoid snapback and high electric field gate oxide breakdown which is because there is no thin gate oxide and it is the major cause for failure in MOSFETs [15][16]. In MESFETS the major cause of its breakdown is due to tunneling and avalanche ionization. As soft breakdown reaches in MESFESTs, the surface electric field becomes large and the barrier height becomes lower at gate thus allowing the electrons to tunnel from the gate metal to the channel [17][18]. By increasing *LaS* and *LaD* lengths the electric fields can be reduced at gate to drain and from gate to source which will increase the breakdown capability of the MESFESTs [19].

III. Modeling and Simulation of Mesfet

This section deals with the modeling and designing of our device and shows all the necessary simulations steps which will prove that our device is very efficient and unique.

In this design the transistor is biased at drain voltage V_{dd}=5V, drain current I_d=120mA and gate voltage of 0.52V. The drain characteristics and power consumption of the transistor are shown in Fig.2 and Fig.3 respectively. The device consumes 0.603 Watt power and has a very reasonable drain voltage. These parameters make it very reasonable choice for power amplifier design [20][21].The small signal model of MESFET is shown in Fig.4. In this model we have defined all intrinsic and extrinsic parameters of MESFET [25].

In this model

R_g is Gate Resistance

L_d is Gate Inductance

R_s is Source Resistance

L_s is Source Inductance

R_d is Drain Resistance

L_d is Drain Inductance

R_i is input Resistance

R_o is output Resistance

C_{ds} is Drain-Source Capacitance

C_{gd} is Gate-Drain Capacitance

C_{sd} is Source- Drain Capacitance

IV. Circuit Analysis and Design

The amplifier is designed and simulated using Agilent ADS software, there are two transistors used in the design, the first transistor is setup in the common source configuration and coupled with a source follower MESFET. The reason to choose common source configuration for MESFET is that the common source configuration increases the gain of the MESFET while reducing the output noise of the amplifier. Following are the desired specifications of the amplifier.

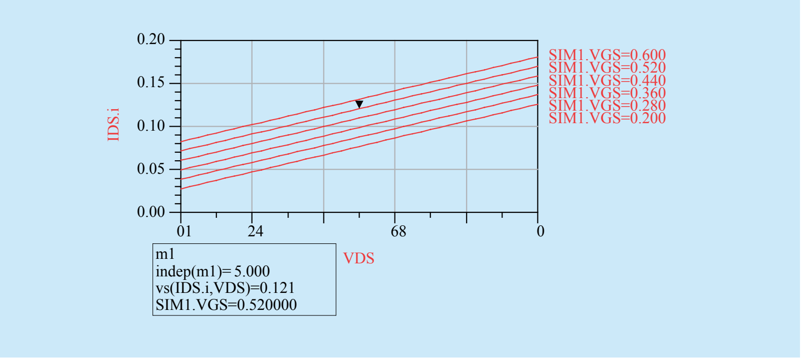

Fig. 2 *Drain current versus bias curves*

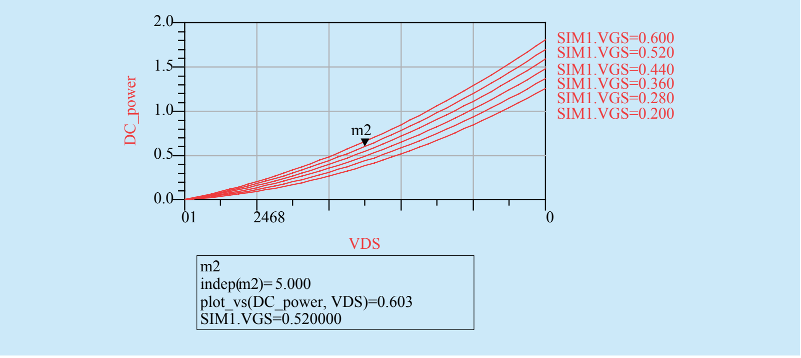

Fig. 3 *DC power consumption versus bias*

The stability Factor is considered the most important design parameter, its value should be more than 1 otherwise the amplifier will generate oscillations. The stability factor is denoted by K for the stability of amplifier following conditions must be met [22][23].

Table I *Amplifier Specifications*

Parameter	Desired Specification
Stability Factor	>1
Gain	More than 20 dB
Input Return Loss	More than 15 dB
Output Return Loss	More than 15 dB
Output Power	More than 20 dBm
Gain	More than 20 dB

$K>1$

$|\Delta|<1$

Where

$$K = \frac{1-|S_{11}|^2-|S_{22}|^2+|\Delta|^2}{2\,|S_{12}S_{21}|} \qquad (1)$$

$$\Delta = S_{11}S_{22} - S_{11}S_{22} \qquad (2)$$

The DC biasing network is not included in this schematic because it is an S-parameter based simulation. The optimized circuit is shown in Fig. 5

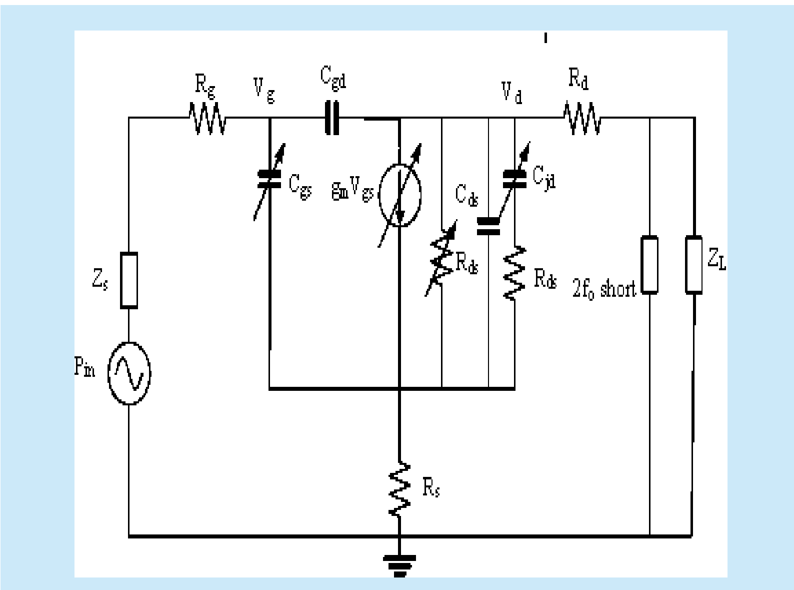

Fig. 4 *Small signal model of MESFET*

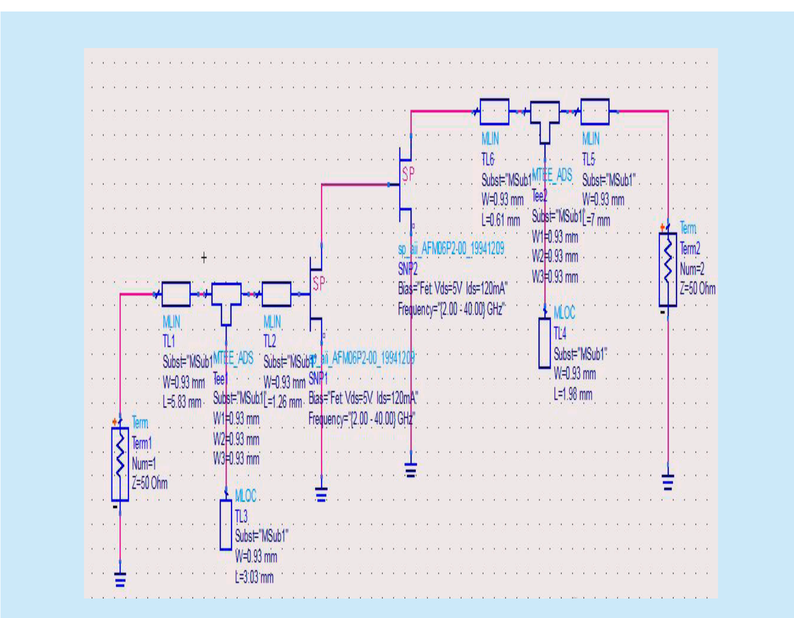

Fig. 5 *Final Design of amplifier*

V. Simulation Results

5.1 Stability factor

The stability factor of amplifier is shown in Fig. 6, it can be seen in the figure that the stability factor is more than 1 for the complete range of operation. Stability factor is very important in designing any type of amplifier because if this factor is below then 1, we have to implement some external stability controlling device thus increasing our cost of the device and also in turn it will increase the size of the device[24][25]. Device size is the main consideration in designing any type of power amplifier for wireless communication because now a day's mobile equipment is becoming smaller and smaller so thus the need for better, efficient and small size PA are required operating with maximum efficiency.

5.2 Input and output return losses

The lower values of input and output return losses indicate design of good impedance matching networks. The input and output matching networks are optimized using the Load pull and Source Pull matching techniques. The Load impedance of amplifier is given by

$$R_{OL} = \frac{2V_{dd}}{I_{\max}} \qquad (3)$$

If the value of load impedance is less than R_{OL} than the value of power depends on max-

Fig. 4 , 5 need vector diagrams, please redo it with font times new roman 6pt imum current swing. On the other hand if the value of load impedance is more than R_{OL} than power depends on maximum voltage swing of the device. The simulated results for input and output return losses are shown in Fig. 7

5.3 Gain

The amplifier power gain is given by [23].

$$G_P = \frac{\text{Power Delivered to the load}}{\text{Power input to the network}} \quad (4)$$

The transducer power gain is plotted versus fundamental output power dBm and shown in Fig. 8. It can be seen clearly that the amplifier has a transducer power gain of about 25.6 dB

The swing of input output voltage of amplifier is shown in Fig. 9

Whereas the values of output power are following

The graph of Power-Added efficiency (PAE) versus fundamental output power is shown in Fig. 10

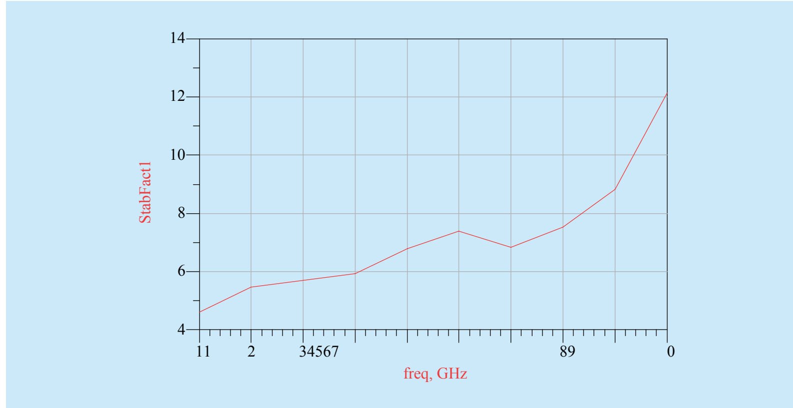

Fig. 6 *Stability Factor*

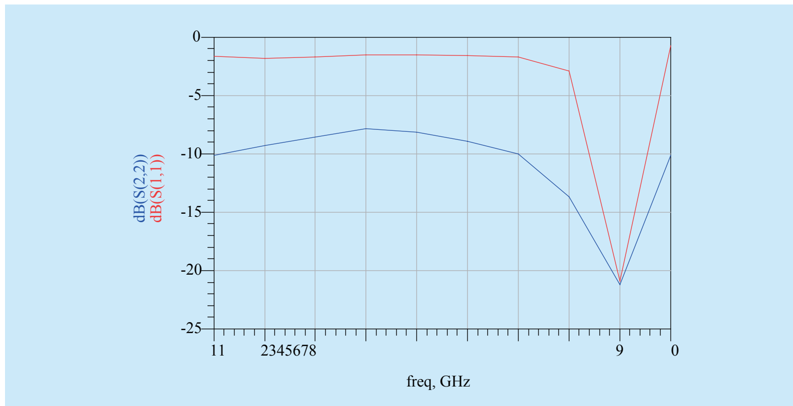

Fig. 7 *Input and output return losses*

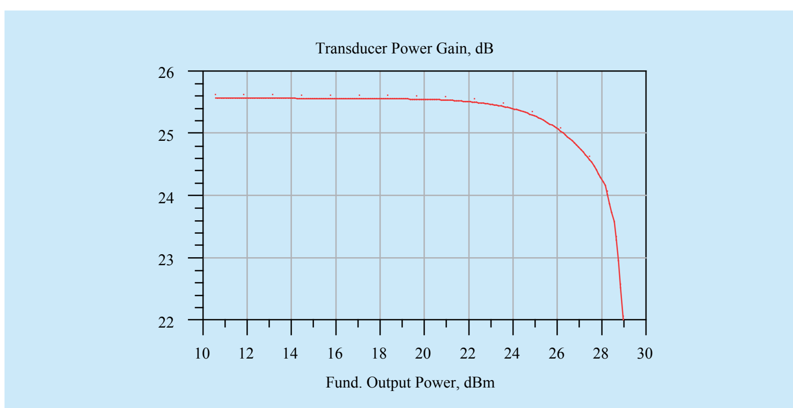

Fig. 8 *Transducer Power Gain versus Output Power*

VI. Comparison

Z. Ma *et. al.* developed a PA using SiGe/Si HBT with lumped passive components, the amplifier has a gain of 8.7 dB, PAE of 30 % and delivers output power trip of 25 dBm [26]. Although the amplifier delivers good amount of output power and PAE but the gain of the amplifier is quite low. In [27], the author developed an 850 mW Xband SiGe power amplifier; he claims to have a gain of 13 dB with an output power of 29.3 dBm. The problem with this design is the non flat nature of gain which is very critical for a next generation communication device. In [28] Campbell *et. al.* designed a compact X band amplifier using GaNFet, the amplifier

has good P_{out} dBm of 30 dBm but the device has non-flat gain. In [29] Chi *et. al.* developed a power amplifier using CMOS, the amplifier achieves a gain of 14.5 dB and delivers output power of 23.8 dBm but the PAE is low for this device. In [30] the amplifier delivers large output power of 27.5 dBm but it has an extremely low gain of 5dB, the PAE is also low. The amplifier suggested in this paper provides a good compromise between the power delivered and the gain with a very good value of peaking PAE. The amplifier shows a flat gain of 27 dB for the entire range of operation with peaking PAE of 30 %. The flatness of gain makes this design a preferable choice for the X-band communication. Furthermore the input

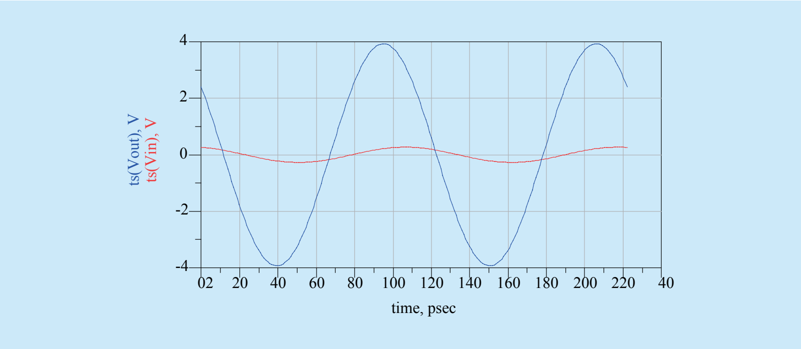

Fig. 9 *Input Output Voltage*

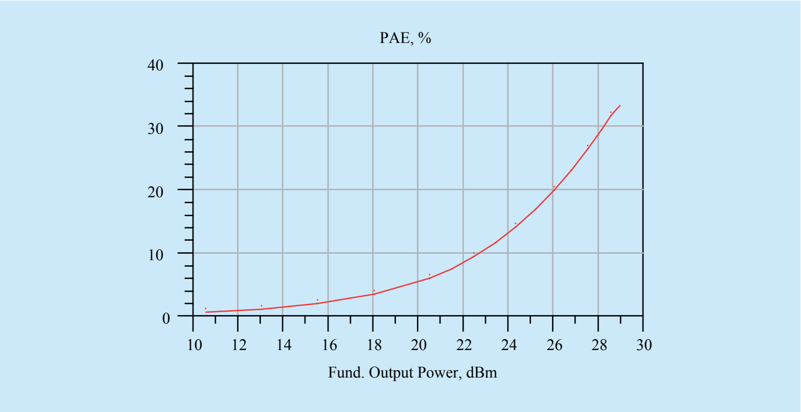

Fig. 10 *Input power versus gain and output power*

Table II *Comparison*

Reference	Gain,dB	Pout, dBm	Peak PAE(%)
This work	25.6	21.894	30
[26]	8.7	25	23
[27]	13	29.3	N/A
[28]	18	30	45
[29]	14.5	23.8	25.8
[30]	5	27.5	17.8

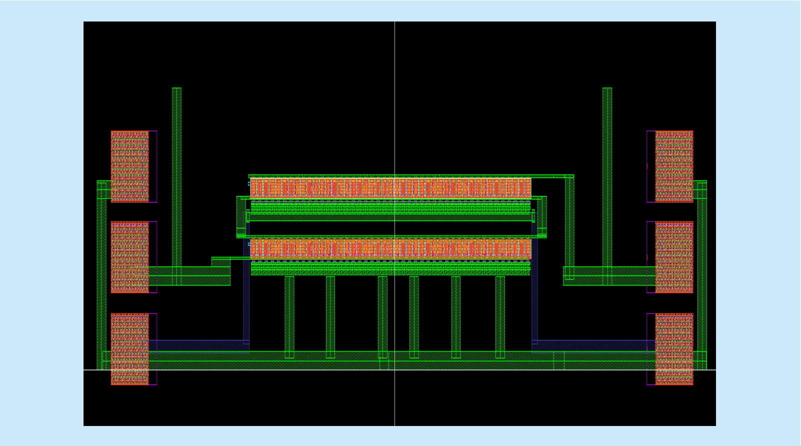

Fig. 11 *Layout of power amplifier*

output return losses of the amplifier are very low. The low values of return losses are result of perfectly matched matching networks. The matching networks are carefully selected and optimized at 9 GHz. The following table shows Gain vs Power output comparison of different research works and our research shows that the PA we designed is unique and has the most gain as compared to the other designs.

VII. Layout of Design

The layout of power amplifier is designed in Cadence. The FR4 material is selected due to its low thermal losses as thermal losses causes decrease in efficiency of a PA. The Layout of Power Amplifier is shown in Fig. 11

VIII. Conclusion

In this work we designed a Class E Power amplifier for X Band operation. The device is suitable for next generation communication devices like 3GPP LTE and WiMax. The device has comparibility better prformance than the recently designed amplifiers interms of gain, power delivered and Peak PAE. The design is optimized for 9 GHz frequency of operation. The amplifier is designed using Microstrip Lines to reduce the size. The Microstrip Lines are non-radiating and do not shows a complex impedance behavior like inductors and capacitors. The design has a high gain of about 25.6 dB, the PAE has peaking value of 30 %.The values of input and output return losses are about 22 dB which shows the accurately designed impedance matching network.

Note

In this paper we have design and analyzed X-band Class E MESFET based Power Amplifier. We have designed and analyzed the PA and showed that our design is more efficient as compared to previous design as shown in the simulated results.

ACKNOWLEDGEMENTS

The Author would like to thank Prof. Zhang Hongxin for his guidance and support during this research work. All the authors have contributed equally to this research work. This work is supported by the National Natural Science Foundation of China (Grant no. 61571063, 61472357, 61501100).

References

[1] U. Varshney, "4G Wireless networks", IT Professional, IEEE Computer Society, vol. 14, October, 2012.

[2] Lepkowski,W.; Wilk,S.J.; Kam,J.; Thornton,T. J."40v MESFETs fabricated on 32 nm SOI CMOS" Custom Integrated Circuits Conference (CICC), Pages: 1 – 4, 2013.

[3] Datta, K.; Hashemi, H. "Performance Limits, Design and Implementation of mm-Wave SiGe HBTClass-E and Stacked Class-E Power Amplifiers." IEEE journal of Solid-State Circuits, Volume: 49, Issue: 10 Pages: 2150 – 2171, 2014.

[4] A. J. Wilkinson and J. K. A. Everard, "Transmission-Line LoadNetwork Topology for Class-E Power Amplifiers," IEEE Trans. Microwave Theory & Tech., vol. 49, no. 6, pp. 1202-1210, Jun. 2000.

[5] W. H. Cantrell, "Tuning Analysis for the High-Q Class-E Power Amplifier," IEEE Trans. Microwave Theory & Tech., vol. 48, no. 12, pp. 2397-2402, Dec. 2000.

[6] Pramod Sharma ; GLA Univ., Mathura, India ; Ravi Prakashdwivedi, "Novel Technique Developed to Design and Simulate RF Amplifier Using GaAs MESFET In ADS Tool", Computational Intelligence and Communication Networks (CICN), 2013 5th international confrence, Pages: 436 – 441.

[7] ZhengZhong, Student Member, IEEE, Yong-XinGuo, Senior Member, IEEE, IEEE Transactions on Microwave Theory and Techniques, VOL. 59,NO. 9, September 2011.

[8] Dr. M. S. Abdul-Wahab, TalibMahmood Ali, Simulation of 9-11GHz High Power Amplifier, Journal of Engineering and Development, Vol. 13, No. 1, March (2009).

[9] Wen-Lin Jung and Jung-Hui Chiu, Stable Broadband Microwave Amplifier Design. IEEE Transactions on Microwave Theory and Techniques, VOL. 41, No. 2, February1993

[10] Sahu, S.; Mohapatra, M.; Panda, A.K. "Dual Gate field plated GaN HEMT With improved breakdown characteristic;" Control, Instrumentation, Communication and Computational Technologies (ICCICCT), 2014 International Conference. Pages: 901 - 905, Year: 2014.

[11] Tombak, et al., "High-efficiency cellular power amplifiers based ona modified LDMOS process on bulk silicon and silicon-on-insulatorsubstrates with integrated power management circuitry," IEEE Trans. Microw. Theory Tech., vol. 60, no. 6, pp. 1862–1869, Jun. 2012.

[12] W. Lepkowski, et al, "Scaling SOI MESFETs to 150 nm CMOS Technologies," IEEE Trans. on Electron Devices, vol. 58, no. 6, pp. 1628-1634, June 2011.

[13] J. Yang , A. Balijepalli , T. J. Thornton , J. Vandersand , B. J. Blalock , M. Mojarradi and M. E. Wood , "Silicon-based integrated MOSFETs and MESFETs: A new paradigm for low power, mixed signal, monolithic systems using commercially available SOI" , Int. J. High Speed Electron. Syst., vol. 16 , no. 2 , pp.723 -732 , 2006.

[14] J. Ervin, et al, "CMOS-Compatible SOI MESFETs With High Breakdown Voltage," IEEE Trans. Electron Devices, vol. 53, pp. 3129- 3135, 2006.

[15] Ghajar, M.R.; Wilk, S.J.; Lepkowski, W.; Bakkaloglu, B.; Thornton, T.J. "Wideband class AB RF power amplifier using CMOS compatible SOI-MESFET device on 150nm technology," Power Amplifiers for Wireless and Radio Applications (PAWR), 2012 IEEE Topical Conference, Pages: 21-24, 2012.

[16] Sharma, P.; Prakashdwivedi, R. "Novel Technique Developed to Design and Simulate RF Amplifier Using GaAs MESFET in ADS Tool." Computational Intelligence and Communication Networks (CICN), 2013 5th International Conference, Pages 436-441. 2013

[17] ZhengZhong, Student Member, IEEE, Yong-XinGuo, Senior Member, IEEE, IEEE Transactions on Microwave Theory and Techniques, VOL. 59, NO. 9, September 2011.

[18] S. J. Wilk et al., "Characterization and modeling of enhanced voltage RF MESFETs on 45nm CMOS for RF applications," IEEE Radio Frequency Integrated Circuits Symposium (RFIC), 2012, pp.413-416, 17-19 June 2012.

[19] Agilent ADS TOM3 documentation, online 2012.

[20] N.M. Memon, M.M. Ahmed, F. Rehman, "A comprehensive four parameters I-V model for GaAs MESFET output characteristics," Solid-state Electronics, Vol.51, pp. 511-516, 2007.

[21] S. J. Wilk et al., "32 dBm Power Amplifier on 45 nm SOI CMOS," IEEE Microwave and Components Letters, vol.23, no.3, pp.161-163, 2013.

[22] B. Francois and P. Reynaert , "A fully integrated watt-level linear 900-MHz CMOS RF power amplifier for LTE-applications" , IEEE Trans. Microw. Theory Tech. , vol. 60 , no. 6 , pp.1878 -1885 , 2012

[23] Guillermo Gonzalez "Microwave Transistor Amplifiers analysis and design" Prentice Hall 2nd edition.

[24] Thomas H Lee "The Design of RF CMOS integrated circuits" Cambridge University press, 2004.

[25] Yaser A. Khalaf, " Systematic Optimization Technique for MESFET Modeling," Ph.D. thesis, Dept. Electron. Eng., Virginia Polytechnic Institute and State Univ., Blacksburg, Virginia, July 2000
[26] Z.Ma, S. Mohammadi, L.H. Lu, P. Bhattacharya, L. P. B. Katehi, S. A. Alterivitz, and G.E. Ponchak, "An X-band high-power amplifier using SiGe/Si HBT and lumped passive components," IEEE Microw. Wireless Compon. Lett., vol. 11, no. 7, pp. 287-289, Jul. 2001.
[27] J. Andrews, J. D. Cressler, W.-M. L. Kuo, and C. Grens, "An 850 mW Xband SiGe power amplifier," in Bipolar/BiCMOS Circuits and Technology Meeting, Oct. 2008, pp. 109–112.
[28] Campbell C.F, Richardson, Poulton "Compact highly integrated X-band power amplifier using commercially available discrete GaN FETs" Asia Pacific Microwave Conference pp. 243-246, 2011
[29] P-S. Chi, Z-M. Tsai, J-L. Kuo, K-Y. Lin, and H. Wang, "An X-band, 23.8-dBm fully integrated power amplifier with 25.8% PAE in 0.18-μm CMOS technology," IEEE Europe Microwave Conference, pp. 1678-1681, Sep. 2010
[30] S. R. Helmi, J-H. Chen, and S. Mohammadi, "A X to Ka-band fully integrated stacked power amplifier in 45 nm CMOS SOI technology," IEEE Silicon Monolithic Integrated Circuits in RF Systems (SiRF), pp. 74-76, Jan. 2014.

Biographies

Muhammad Saad Khan, was born in 1984 in Pakistan. He received his M.Sc Electrical Engineering Degree in 2010 from Blekinge Institute of Technology, Karlskrona, Sweden and BS.c Electrical Engineering Degree in 2007 from Bahauddin Zakariya University Multan, Pakistan. He has been working as a Lecturer in Electrical Engineering Department BahauddinZakariya University Pakistan. Currently he is percuing his Ph.D Degree at the School of Electronic Engineering, Beijing University of Posts and Telecommunications, Beijing 100876, China. His research interests are VLSI design, Microwave and antenna Propagation, RFIC design, Wireless and Optical communications, Low noise amplifiers and Power Amplifiers. Email: saadkhan9@gmail.com

Hongxin Zhang, was born in 1967 in Shandong province of China.He is the professor of Electrical Engineering Institute of Beijing university of posts and telecommunications, and also the Doctoral tutor.He is the director of the Broadband communications and microwave technology center. He is the review expert of the national natural science fund project. He is also the evaluation expert of Education degree and graduate education development center. Email: hongxinzhang@bupt.edu.cn.

Fan Zhang, was born in 1978, Zhejiang, China, acquired his Ph.D degree from department of computer science and engineering in University of Connecticut, America. Now he is a teacher of college of information and electrical engineering, Zhejiang University, and his interesting research filed includes information security, computer architecture, human-computer interaction, and sensor networks. Email: fanzhang@zju.edu.cn

Sulman Shahzad, received the BSc in electrical engineering degree from Bahaudin Zakariya University (BZU) Multan in 2015 While with BZU, he studied RFIC Design for wideband applications. Currently he is working as Protection and Instrumentation Engineer in NTDC Pakistan. He has done many online courses from the institutions like MIT, Georgia tech. His research interests are RFIC Design, VLSI Design, milimeter waves circuits and Transceiver design. Email: salmanshahzad05@gmail.com

Rahat Ullah, received his M.Sc Electronics degree(2008) from University of Peshawar Pakistan and MS (CS) Telecommunication and Networking degree(2012) from Gandhara University Peshawar Pakistan. He worked as a lecturer/Assistant professor in Sarhad University Peshawar Pakistan for 6 years. Currently he is working toward his PhD from State Key Laboratory of Information Photonics and Optical Communications, Beijing University of Posts and telecommunications, China. His research interests are optical communications, passive optical networks, optical multicarriers generation and radio over fiber, RFIC Design, Transducer Design and Power Amplifiers. Email: r.rahatullah@gmail.com

Sajid Ali, received the PhD degree and Postdoctoral from College of Information Science and Technology, Beijing Normal University, Engineering Research Center of Virtual Reality and Application, Ministry of Education, Beijing, China, and the MSCS and MSc (CS) degrees from Department of

Computer Science, University of Agriculture, Faisalabad, Pakistan. Currently he is a faculty member the department of computer Science, University of Education, Lahore, Pakistan. His current research interests include biometrics, motion retarget animation, information system and computer network. Email: snfa_bnu@yahoo.com

Qasim Ali Arain, is a Ph.D. candidate at the school of Electronic Engineering, Beijing University of Posts & Telecommunications Beijing China. He is currently working as an Assistant Professor in Department of Software Engineering Mehran University of Engineering and Technology, Jamshoro, Pakistan. He has completed his Bachelor's in software Engineering from Mehran UET Jamshoro in 2005 and completed his masters in Information Technology in 2010. He is a Cisco certified Academy Instructor (CCAI), Cisco security Specialist, Microsoft certified System Engineer (MCSE), Sun certified Java Programmer (SCJP). His current research interest includes Location based services, Network Security and Indoor Positioning. Email: Qasim.arain@faculty.muet.edu.pk

Manzoor Ahmed, is currently working as Post doctorate researcher at the department of Electronic Engineering, Tsinghua University, Beijing, China. He has completed the Ph.D. degree from the Beijing University of Posts and Telecommunications, Beijing, China, in 2015. He received the M.Phil and B.E. degree from Pakistan. He served NTC and Baluchistan engineering university Khuzdar for 12 and 3 years respectively. His research interests include the non-cooperative and cooperative game theoretic based resource management in hierarchical heterogeneous networks, interference management in small cell networks, D2D communication and 5G networks, physical layer security and information theoretic security. Email: manzoor.achakzai@gmail.com